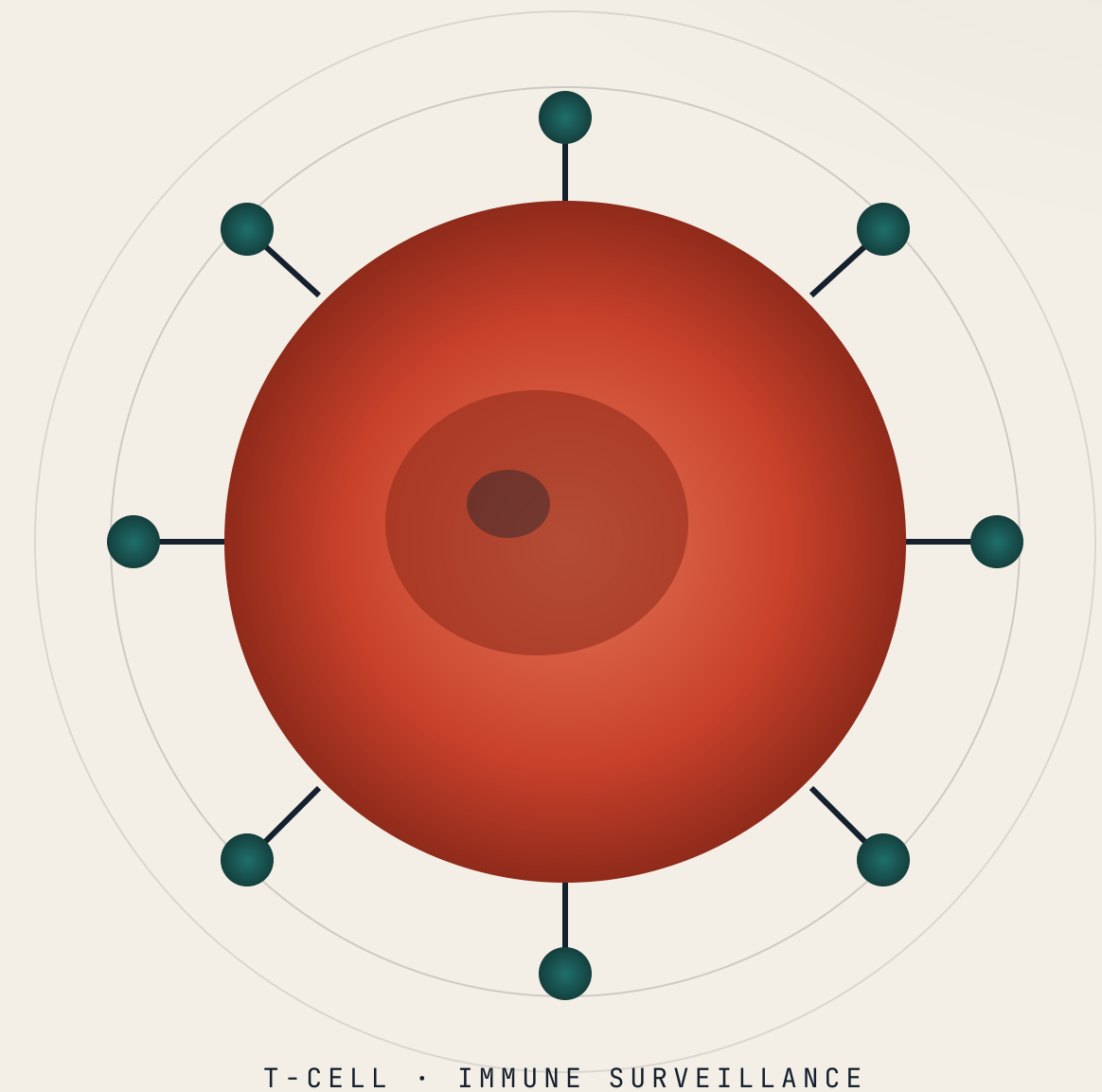


Immunity, *Limits*, and the Energy that Drives Them.

Course GSM • Friedrich-Schiller-Universität
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Format 10-slide summary



Three systems, one body — what we'll cover today.

01

Immunity & Inflammation

How the body protects integrity through innate & adaptive defence — and how exercise tunes the inflammatory thermostat.

02

Limits of Performance

Definitions, determinants and the age-dependent ceiling on what muscular work the organism can deliver.

03

Energy Balance

From food to ATP — the energy systems that fuel contraction, and the macronutrients that feed them.

DEFENDING BODILY INTEGRITY

The immune system runs *two parallel programs* — one fast and unspecific, one slow and remembered.

FIRST LINE • INNATE

Innate Immunity

Non-specific, immediate response — no prior sensitisation required.

COMPONENTS	Skin, mucous membranes, neutrophils, macrophages, NK cells, complement
TRIGGER	Pathogen-associated molecular patterns (PAMPs)
SPEED	Activation within minutes

SECOND LINE • ADAPTIVE

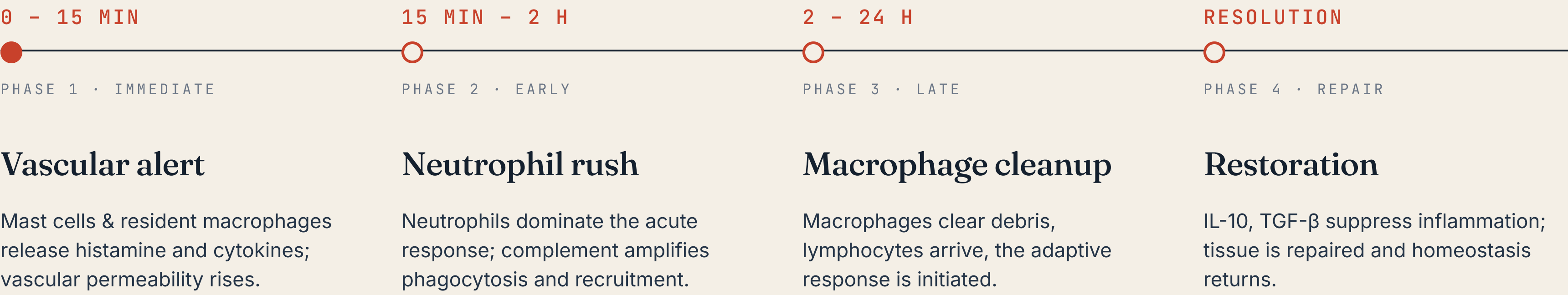
Adaptive Immunity

Antigen-specific, slower to mobilise — but generates memory.

COMPONENTS	T lymphocytes (cellular), B lymphocytes (humoral), antibodies
TRIGGER	Specific antigens after sensitisation
SPEED	Days on first exposure, hours on re-exposure

A SELF-LIMITING CASCADE – WHEN IT WORKS.

Acute inflammation unfolds in *four phases*, each with a different cellular cast.



ACUTE • HOURS TO DAYS	CHRONIC • WEEKS TO YEARS
Protective and self-limiting — fights infection, primes repair.	Detrimental — implicated in CVD, T2D, autoimmunity, neurodegeneration, cancer.

SAME STIMULUS, TWO TIMESCALES

A single bout *mobilises* immune cells; a habit of training *lowers* the inflammatory baseline.

ACUTE BOUT

Mobilisation & *myokine release*

- Lymphocyte & NK-cell trafficking rises transiently in circulation.
- IL-6 from muscle acts as a myokine — anti-inflammatory signalling despite the spike.
- Immune surveillance enhanced as mobilised cells migrate to peripheral tissues.

CHRONIC TRAINING

Lower baseline, *better surveillance*

- Resting IL-6, TNF- α , CRP all reduced.
- T-cell numbers up; vaccination antibody response improved.
- Reduced infection risk in active people — but a **U-shape** appears at extreme endurance loads.

Regular moderate-to-vigorous activity enhances immune competence and dampens chronic inflammation — protecting against age-related decline and lifestyle disease.

PRACTICAL TAKEAWAY • §1 LECTURE NOTES

WHAT DO WE MEAN BY PHYSICAL PERFORMANCE?

Performance is defined by *four overlapping lenses* — biological, biochemical, lifespan and willed.

BIOLOGICAL-PHYSICAL

Muscular work per unit of time.

The sum of all determinants currently available to deliver mechanical body work.

$$P = \text{work} / \text{time}$$

LIFESPAN PERSPECTIVE

A trajectory, not a constant.

Genetically pre-set, behaviourally shaped — performance is part of biological, psychological and social development across the entire lifespan.

$$\text{capacity}(t) \text{ across the lifespan}$$

BIOCHEMICAL ENERGY SUPPLY

ATP delivery per unit of time.

Aerobic and anaerobic energy provided by high-energy phosphates to fuel neuromuscular work.

$$\text{aerobic} + \text{anaerobic ATP flux}$$

EXTENDED DEFINITION

Value created against a goal.

Depends on biological, psychological and social resources, on internal and external stress — and on the *willingness to perform*.

$$\text{resources} \times \text{stress} \times \text{will}$$

WHAT SETS THE CEILING?

Two *primary determinants* shape capacity — and ten *modifiers* push it up or down.

Primary determinants

01

Biological structure

The anatomical scaffold — at the level of the whole organism and every sub-system within it.

02

Physiological functionality

How well the structure functions — across systems, control circuits and cellular metabolism.

Secondary determinants

- | | |
|-------------------------------|---|
| + Biological ageing processes | + Health as physiological functionality |
| + Disease & dysfunctionality | + Health-related lifestyles |
| + Nutrient supply | + Movement & physical activity |
| + Genetics & epigenetics | + Environmental factors |
| + Sensorimotor & perception | + Psyche & willingness to perform |

ENERGY SUPPLY BY DURATION

Four metabolic systems, each with its own *flow rate* and *time-to-fatigue*.

DURATION	ENERGY SOURCE	EXAMPLES	FLOW RATE · MMOL · KG ⁻¹ · S ⁻¹
< 2 s	Stored ATP	Power lift, javelin throw, golf swing, tennis serve	3.0 – 6.0
≤ 6–8 s	ATP + creatine phosphate (PCr)	Sprints, fast breaks, gymnastics routines	3.0 – 6.0
≤ 90 s	+ Anaerobic glycolysis (lactate)	200–400 m sprints, 100 m swim	1.5 – 3.0
~ 800 m	~50 % aerobic + ~50 % anaerobic	Middle-distance running	0.5 – 0.75
hours	Aerobic glycolysis → β-oxidation	Long-distance running, endurance events	0.24 – 0.40

"The faster you run, the sooner you get tired" — speed buys flow rate at the cost of capacity.

Capacities · ATP+PCr ≈ 19 mmol·g⁻¹ (≤8 s) · anaerobic glycolysis ≈ 50 mmol·g⁻¹ (≤60 s) · aerobic pathway limited only by substrate (hours – days).

IN = OUT – BUT ONLY ~40 % SURVIVES THE TRIP.

Macronutrients carry the energy; *ATP is the currency* the cell will accept.

80 · 10¹² cells

All require ATP to maintain structure and function.

< 40 %

Of food energy is available as utilisable / stored energy for skeletal muscle work.

689 kcal · ~34 % → ATP

Liberated by full oxidation of 1 mol glucose; about a third is conserved in ATP bonds.

FAT

9.3 kcal·g⁻¹

Largest store of biological energy. 50–70 % of fuel at low-to-moderate intensity; > 80 % in prolonged exercise.

1 triacylglycerol → ~460 ATP

CARBOHYDRATE

4.1 kcal·g⁻¹

Primary fuel for anaerobic work and the CNS. Spares protein, primes fat catabolism. 60–70 % of intake for hard-training athletes.

1 glucose → net 32 ATP

PROTEIN

4.1 kcal·g⁻¹

Structural, not preferred fuel. RDA 0.83 g·kg⁻¹; athletes 1.2–1.8 g·kg⁻¹. Alanine cycle supplies ≤ 45 % of liver glucose in long efforts.

8 of 20 amino acids are essential

WHERE THE THREE THREADS MEET

Immunity, performance and energy are *one regulated system* — measured in ATP, modulated by movement.

01 • Immunity

Acute inflammation protects; chronic inflammation harms.

Regular activity tunes the inflammatory thermostat downward — IL-6 from muscle is a myokine, not just a stress signal.

02 • Performance

Capacity is biological structure × functional state × will.

Two primary determinants, ten secondary modifiers, and an age-dependent ceiling shape what the organism can deliver.

03 • Energy

Flow rate trades against capacity.

ATP-PCr pays out fast and runs dry; β-oxidation pays slowly but pays for hours. Intensity picks the system, duration empties it.

Next: *Cardiovascular & respiratory responses to exercise* — how the delivery infrastructure scales to meet the metabolic bill.